ObjDesMap

Short object descriptions for the Map Explorer — condensed from StoModInf. Objects listed in the same order as `assets/js/data.js`.

MOON

Lunar Module "Eagle" descent stage · Apollo 11 · 1969

On July 20, 1969, Eagle carried Neil Armstrong and Buzz Aldrin to the first human landing on the Moon, then served as the launch platform that sent them home. Its lower half still stands in the Sea of Tranquility, next to a laser mirror that scientists are still using today.

Surveyor 3 lander · Surveyor 3 + Apollo 12 · 1967

This robotic scout bounced twice when its engine failed to shut off, then dug trenches proving the lunar soil could hold a crewed lander. Two and a half years later, Apollo 12 astronauts walked over to it — the first time humans ever visited a robot on another world — and took its camera and scoop back to Earth.

Golf balls · Apollo 14 · 1971

On February 6, 1971, Alan Shepard fixed a six-iron head to a lunar sampling tool and swung one-handed, his bulky spacesuit allowing nothing more. He joked one ball flew "miles and miles and miles" — image analysis later measured about 24 and 40 yards — making golf the only sport ever played on the Moon.

Laser Ranging Retroreflector · Apollo 15 · 1971

A suitcase-sized array of 300 mirrors that needs no batteries, solar panels or moving parts. Laser pulses fired from Earth bounce back in about 2.5 seconds, measuring the Moon's distance to the millimetre and revealing that it drifts about 3.8 cm farther away each year — measurements still being recorded in 2026.

Family photograph · Apollo 16 · 1972

On his final moonwalk, Charlie Duke laid a plastic-wrapped portrait of his wife and two sons on the lunar dust, signed on the back: "This is the family of Astronaut Duke from Planet Earth. Landed on the Moon, April 1972." With no scientific purpose at all, it is one of the most human objects ever left on another world.

Lunar Roving Vehicle · Apollo 17 · 1972

The last rover ever driven on the Moon covered about 36 km and helped Cernan and Schmitt collect 110.5 kg of samples, including orange volcanic glass from Shorty Crater. It is still parked about 100 m from the lander, its camera facing the spot where the crew lifted off for home.

Waste containment bags · Apollo 11 · 1969

To save weight for the precious Moon rocks, the first moonwalkers left their sealed bags of human waste behind at Tranquility Base. The least glamorous artefact on the Moon may yet prove useful: scientists have proposed studying such bags to learn whether microbes can survive decades on the lunar surface.

Not covered in StoModInf — written from general knowledge; please verify before publishing.

MARS

Viking 1 lander · 1976

The first successful U.S. Mars lander sent back the first photographs ever taken from the Martian surface and, built for 90 days, kept working for more than six years. Its air readings later proved that certain meteorites on Earth came from Mars — until a faulty software update in 1982 turned its antenna away from home.

Sojourner rover · Mars Pathfinder · 1997

Small enough to carry in your arms, Sojourner became the first wheeled vehicle to drive on another planet when it landed in Ares Vallis. Designed for just seven Martian days, it worked for 83, studying rocks and paving the way for every Mars rover since.

Spirit rover · MER-A · 2004

Spirit's biggest discovery was an accident: dragging a broken front wheel through the dirt, it uncovered a mineral that forms around hot springs — proof that Mars was once warm and wet. Built for 90 days, it explored Gusev Crater for over six years before getting stuck in a sand trap engineers nicknamed "Troy".

Opportunity rover · MER-B · 2004

Meant to last three months, Opportunity explored for nearly fifteen years and drove more than 45 km — farther than any rover before it. It found tiny hematite-rich spheres nicknamed "Martian Blueberries", strong evidence of ancient water, until a planet-wide dust storm cut its power in 2018.

Phoenix lander · 2008

Built partly from spare parts and named after the bird that rises from the ashes, Phoenix landed closer to Mars's north pole than any spacecraft before. It confirmed water ice beneath the surface and even watched snow falling from clouds about four kilometres overhead, before the arctic winter silenced it.

InSight lander · 2018

Instead of driving around, InSight stayed put and acted like a doctor checking Mars's vital signs. It recorded the first marsquakes ever detected — more than 1,300 seismic events — and revealed that Mars has a molten core, until dust slowly smothered its solar panels in 2022.

Ingenuity Mars Helicopter · 2021

On April 19, 2021, this 1.8 kg, softball-sized helicopter made the first controlled flight on another planet, its blades spinning about five times faster than an Earth helicopter's to grip the thin air. Planned for five test flights, it flew 72 — carrying a tiny scrap of fabric from the Wright brothers' first airplane.